

MICROFI COLLECT

Test Documentation

IEEE 829–Style Verification & Validation Pack

Test Plan · Design · Cases · Procedures
Transmittal · Log · Incidents · Summary

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Design system aligned with Business Analysis v3.1

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1. Document Map

This pack follows the classical IEEE 829 test documentation set, adapted to MICROFI Collect (MVP). Each component below is a full section of this document.

#	Component	Purpose in this pack
1	Test Plan	Approach, scope, resources, schedule, deliverables
2	Test Design Specification	Features, conditions, coverage per module
3	Test Case Specification	Step-by-step cases with expected results
4	Test Procedure Specification	Ordered execution of related case groups
5	Test Item Transmittal Report	Handoff of builds from development to test
6	Test Log	Chronological execution record
7	Test Incident Report	Deviations / defects for investigation
8	Test Summary Report	Overall assessment for stakeholders

References: Business Analysis & Requirements; System Architecture & Software Design; Offline-First Geolocation Spec; OpenAPI / Swagger.

2. Test Plan

Identifier

TP-MICROFI-V1.0 — Test Plan for MICROFI Collect Minimum Viable Product.

Introduction

MICROFI is a hybrid offline-first platform for itinerant microfinance collection in the CEMAC zone. This plan governs verification of MVP flagships: secure agent terminal, escrow lockout, offline collection with GPS gate and denomination, digital OFJ, and client read views.

Test items

- Flutter agent application (Android)
- Flutter client application (read profile/balance)
- Next.js back office (enrollment, OFJ, ceilings)
- Core Backend (Spring Boot 4) and public API via Gateway
- CBS Middleware mock
- Docker infrastructure: PostgreSQL, Redis, RabbitMQ

Features to be tested (in scope)

FR-01, FR-02, FR-04, FR-06, FR-07, FR-08, FR-10, FR-12, FR-16, FR-17, FR-18 (stub), FR-20, FR-21; BR-01, BR-02, BR-03, BR-05; NFR-01, NFR-08, NFR-09 (pulse behaviour).

Features not to be tested (V1.0)

Deep multi-bank GIMAC automation (FR-03 full); production CBS vendor certification; national-scale load; iOS; production USSD gateway.

Approach

1. **Unit** — validators (escrow, denomination, GPS presence, JWT).
2. **Integration** — Core with Postgres/Redis/RabbitMQ; sync and middleware mock.
3. **API** — Swagger UI / OpenAPI (springdoc); positive and negative cases.
4. **System / E2E** — physical Android preferred for GPS and offline.
5. **Acceptance** — scripted demo (Section 5.8).

Item pass / fail criteria

- **Pass:** Expected result observed; no Critical defect on money path or GPS gate.
- **Fail:** Deviation from expected; logged as incident (Section 8).
- **Blocked:** Environment or dependency prevents execution; recorded in Test Log.

Suspension / resumption

Suspend formal acceptance if Critical defect on escrow, GPS gate, or double-posting remains open. Resume after fix is transmitted (Section 6) and smoke re-run.

Test deliverables

- This document (plan, design, cases, procedures, templates)
- Executed Test Log and Incident Reports
- Test Summary Report at MVP freeze
- Swagger export / OpenAPI snapshot used for the run

Environmental needs

Docker Compose stack; JDK 21; Flutter 3.x; Node 20; physical Android with GPS; optional Bluetooth printer.

Responsibilities

Role	Responsibility
Author / developer	Build features, unit/integration tests, defect fixes
Tester / author	Execute procedures, log results, file incidents
Supervisor	Accept demo; approve summary report
MFI liaison (optional)	Operational realism of OFJ and collection

Staffing and training

Solo or small academic team; training = this pack + BA use cases + Swagger walkthrough.

Schedule (indicative, aligned to 6-week MVP)

Phase	Test focus	Week
0 Foundation	Health, gateway, migrations	1
1 Auth + escrow	TC-AUTH, TC-ESC	2
2 Collection + GPS	TC-COL	3
3 Offline sync	TC-OFF	4
4 OFJ + client	TC-OFJ, TC-CLI	5
5 Harden + acceptance	TC-GEO, TC-SEC, demo procedure	6

Risks and contingencies

GPS / emulator

Use physical devices for all P0 COL/GEO cases.

CBS mock only

Vendor UAT scheduled when credentials exist; not a V1.0 exit blocker.

Performance & accessibility (future)

National-scale load and formal accessibility audits are out of scope for V1.0 formal tests. Track as V1.1+ work: API latency budgets under sync bursts; WCAG-oriented review of back office.

Approvals

Author: _____ Supervisor: _____ Date: _____

3. Test Design Specification

Identifier

TDS-MICROFI-V1.0

Features to be tested (design view)

Testing is organised by functional module. Each feature is refined into **test conditions** (what must be true) and **coverage notes**.

AUTH — Authentication & enrollment

- **Conditions:** Valid PIN+IMEI; invalid PIN; IMEI mismatch; admin can enroll.
- **Coverage:** FR-01, FR-02, UC-01, UC-02.
- **Cases:** TC-AUTH-01..04.

ESC — Escrow & logout

- **Conditions:** Under ceiling; at/over ceiling; offline uses last synced remaining; top-up restores; override (P1).
- **Coverage:** FR-04, BR-03, UC-04.
- **Cases:** TC-ESC-01..05.

COL — Collection, denomination, GPS gate

- **Conditions:** Happy path; GPS off; bad accuracy; denomination mismatch; client ID methods; immutable geotag.
- **Coverage:** FR-06, FR-08, FR-12, BR-02, BR-05.
- **Cases:** TC-COL-01..06.

OFF — Offline & sync

- **Conditions:** Encrypted queue; idempotent sync; multi-tx burst; failed sync retains queue.
- **Coverage:** FR-07, NFR-01, NFR-08.
- **Cases:** TC-OFF-01..04.

GEO — Continuous tracking

- **Conditions:** 5-min pulse offline; batch 202; map last-known; off-duty stop (P1).
- **Coverage:** FR-10, FR-11, NFR-09, NFR-10.
- **Cases:** TC-GEO-01..04.

OFJ — End of day

- **Conditions:** $\Delta = 0$ close; negative variance; positive per policy; export stub.
- **Coverage:** FR-16, FR-17, FR-18, BR-01.
- **Cases:** TC-OFJ-01..04.

CLI — Client

- **Conditions:** Own profile only; balance after sync; history (P1).
- **Coverage:** FR-20, FR-21, FR-22.
- **Cases:** TC-CLI-01..03.

SEC — Security & gateway

- **Conditions:** No JWT; branch isolation; rate limit; encrypted offline DB.
- **Coverage:** NFR-01, STRIDE (Elevation, Disclosure, DoS).
- **Cases:** TC-SEC-01..04.

Approach refinements

Negative testing is mandatory on money path and GPS gate. Offline scenarios use airplane mode. API exploration and contract checks use **Swagger UI** / **OpenAPI (springdoc)**.

Feature pass criteria

A feature passes when all linked **P0** cases Pass and no open Critical incident remains for that feature.

4. Test Case Specification

Convention. TC-MODULE-NN. Every case below includes all five fields: **Objective**, **Preconditions**, **Steps**, **Expected**, **Refs**.

Prioritization method

Priorities are **risk-based**, aligned with the BA Feature Priority Matrix and money-path safety:

- **P0 — MVP blocker:** Failure allows incorrect money movement, double posting, lockout bypass, collection without GPS, or blocks the acceptance demo. Must Pass before freeze.
- **P1 — Fast-Follow:** Important for pilot quality (overrides, history, rate limit, map polish) but not required for V1.0 academic acceptance.
- **P2 — Backlog:** Deferred with FR backlog (full multi-bank, advanced analytics).

Coverage metrics (catalogue size)

Category	Count	Notes
Total test cases in catalogue	38	Including boundary & usability
P0	28	MVP exit set
P1	10	Fast-Follow
P2	0	None in this pack
Planned automation (V1)	≈40% of P0	Unit/API via JUnit + Swagger; device cases manual

Seed data (shared preconditions)

Branch BR-YDE-01; agents A1 (ceiling 100,000 XAF), A2 (50,000 XAF); clients C01–C10; BEAC denomination grid. No production PII.

TC-AUTH

TC-AUTH-01 P0

Agent login success

Objective: Valid agent authenticates on bound device.

Preconditions: Enrolled; IMEI matches; PIN correct.

Steps: Open app → enter PIN → submit.

Expected: JWT; home screen; secure session storage.

Refs: FR-01, UC-01.

TC-AUTH-02 P0

Wrong PIN rejected

Steps: Submit incorrect PIN (up to limit).

Expected: Denied; no JWT.

Refs: FR-01.

TC-AUTH-03 P0

IMEI mismatch rejected

Preconditions: Bound to IMEI-A; run on IMEI-B.

Expected: Login refused.

Refs: FR-01.

TC-AUTH-04 P0

Admin enrolls agent

Steps: Back office: create agent, branch, IMEI, PIN/escrow.

Expected: Agent logs in on matching device.

Refs: FR-02, UC-02.

TC-ESC**TC-ESC-01 P0****Collection under ceiling**

Preconditions: Remaining escrow > amount.

Steps: Valid collection (GPS on, denomination OK).

Expected: Success; remaining decreased.

Refs: FR-04, BR-03.

TC-ESC-02 P0**Lockout at ceiling**

Steps: Attempt collection when remaining insufficient.

Expected: Blocked; no new financial row.

Refs: FR-04, BR-03, UC-04.

TC-ESC-03 P0**Offline lockout uses last synced ceiling**

Preconditions: Last synced remaining 5,000 XAF; offline.

Steps: Attempt 6,000 XAF offline.

Expected: Local rejection; not queued as valid.

Refs: FR-04, FR-07.

TC-ESC-04 P1**Top-up restores collection**

Objective: After lockout, crediting escrow allows collection again.

Preconditions: Agent A2 locked out (remaining = 0); cashier/admin can post MVP top-up.

Steps: 1) Confirm lockout. 2) Post top-up of 20,000 XAF on A2. 3) Agent refreshes escrow. 4) Collect 5,000 XAF with GPS on.

Expected: Remaining shows 20,000 then 15,000; collection succeeds.

Refs: FR-03 (MVP path), FR-04.

TC-ESC-05 P1**Temporary ceiling override with audit**

Objective: Only authorised role can raise ceiling; action is audited.

Preconditions: A2 at lockout; user Admin has override right; user Agent does not.

Steps: 1) As Agent, attempt override API/UI — expect deny. 2) As Admin, set temporary ceiling +30,000 XAF with reason. 3) Open audit log.

Expected: Agent denied; Admin succeeds; audit entry with actor, old/new ceiling, timestamp.

Refs: FR-05, UC-05.

TC-ESC-06 P0**Boundary — collection equals remaining escrow**

Objective: Exact ceiling consumption is allowed and leaves remaining = 0.

Preconditions: A1 remaining escrow synchronised to exactly 10,000 XAF; client C01 available.

Steps: 1) Start collection of 10,000 XAF. 2) Enter denomination lines summing to 10,000. 3) GPS on; confirm.

Expected: Transaction accepted; remaining escrow = 0; further collection blocked (as TC-ESC-02).

Refs: FR-04, BR-03.

TC-ESC-07 P0

Boundary — collection exceeds remaining by 1 XAF

Objective: Amount remaining+1 is rejected.

Preconditions: A1 remaining = 10,000 XAF.

Steps: Attempt collection of 10,001 XAF with valid denomination and GPS.

Expected: Rejected before commit; remaining still 10,000; no financial row.

Refs: FR-04, BR-03.

TC-ESC-08 P0

Boundary — remaining–1 still allowed

Objective: Amount one unit under remaining succeeds.

Preconditions: A1 remaining = 10,000 XAF.

Steps: Collect 9,999 XAF; valid denomination; GPS on.

Expected: Success; remaining = 1 XAF.

Refs: FR-04, BR-03.

TC-COL

TC-COL-01 P0

Happy path online collection

Steps: Client ID; amount; denomination sum; GPS on; confirm.

Expected: UTC + lat/lon stored.

Refs: FR-06, FR-08, FR-12, BR-02, BR-05.

TC-COL-02 P0

GPS off blocks collection

Expected: No financial row.

Refs: FR-12, BR-05.

TC-COL-03 P0

Invalid GPS fix rejected

Objective: Collection requires a usable fix (accuracy within policy).

Preconditions: Agent logged in; mock or device returns accuracy > configured threshold (e.g. > 50 m) or null coordinates.

Steps: 1) Start collection. 2) Enter amount and valid denomination. 3) Attempt confirm while fix is invalid.

Expected: Submit blocked or API 422; no financial row until accuracy is acceptable.
Refs: FR-12, BR-05, Geolocation Spec.

TC-COL-04 P0**Denomination sum mismatch**

Steps: Amount 10,000; lines sum 9,500.
Expected: Validation error; no commit.
Refs: FR-08, BR-02.

TC-COL-05 P0**Client identification by member ID and phone**

Objective: Agent can resolve the correct client before amount entry.
Preconditions: Clients C01 (known member ID) and C02 (known phone) in seed data.
Steps: 1) Search by member ID of C01 — select. 2) Clear. 3) Search by phone of C02 — select.
Expected: Correct client name/ID shown in context for each search; no cross-identity mix-up.
Refs: FR-06.

TC-COL-06 P0**Geotag immutable after save**

Objective: Lat/lon recorded at collection cannot be altered by agent or API.
Preconditions: TC-COL-01 completed; note stored coordinates.
Steps: 1) Attempt UI edit of coordinates (if any control exists). 2) Attempt PATCH/update with different lat/lon on collection id.
Expected: UI has no edit; API returns 403/405/422; original geotag unchanged in DB.
Refs: FR-12.

TC-OFF**TC-OFF-01 P0****Offline collection encrypted**

Expected: Local encrypted store; pending indicator.
Refs: FR-07, NFR-01.

TC-OFF-02 P0**Idempotent sync**

Expected: One server row per device_tx_id.
Refs: FR-07, NFR-08.

TC-OFF-03 P0**Offline burst of three collections**

Objective: Multiple offline deposits sync completely once.

Preconditions: Agent online escrow sufficient; then airplane mode.

Steps: 1) Record three valid offline collections (different clients). 2) Restore network. 3) Sync once.

Expected: Three server rows; escrow reduced by sum of three; timestamps ordered.

Refs: FR-07.

TC-OFF-04 P0

Failed sync retains local queue

Objective: Server errors must not drop offline data.

Preconditions: At least one unsynced local collection; ability to mock API 500.

Steps: 1) Trigger sync against failing endpoint. 2) Confirm local pending remains. 3) Restore API; sync again.

Expected: First attempt fails without purge; second attempt posts once (idempotent).

Refs: FR-07, NFR-08.

TC-GEO

TC-GEO-01 P0

Background pulse offline

Expected: $\approx$ 5-min points; encrypted.

Refs: FR-10, NFR-09.

TC-GEO-02 P0

Batch location sync 202

Refs: FR-10.

TC-GEO-03 P1

Map shows last known during blackout

Objective: Manager UI must not invent movement while agent is offline.

Preconditions: Agent had at least one online ping; then offline $\geq$ 10 minutes with local pulses.

Steps: 1) Open back-office map during blackout. 2) Note marker position. 3) After sync, re-open map.

Expected: During blackout, marker at last online point only; after sync, FR-11 polyline includes offline milestones.

Refs: FR-11, Geolocation Spec.

TC-GEO-04 P1

Tracking stops outside service hours

Objective: No new trail points after configured end of service window.

Preconditions: Branch window ends at 18:00 WAT; agent session was active.

Steps: 1) Set device clock (or config) past end of window. 2) Wait for two pulse intervals. 3) Inspect local tracking table.

Expected: No new rows after window end; optional banner that tracking is off-duty.

Refs: NFR-10.

TC-OFJ

TC-OFJ-01 P0

Balanced close

Expected: $\Delta = 0$; export enqueued.

Refs: FR-16, FR-18, BR-01.

TC-OFJ-02 P0

Negative variance blocks close

Objective: Physical count below system total cannot be ignored.

Preconditions: System total S known; agent/cashier role on OFJ screen.

Steps: 1) Enter physical count $S - 1,000$. 2) Attempt close.

Expected: Close refused or investigation ticket forced; agent debt recorded; audit entry; no silent success.

Refs: FR-16, FR-17, BR-01.

TC-OFJ-03 P0

Positive variance handled per policy

Objective: Count above system total follows documented business rule.

Preconditions: System total S .

Steps: Enter physical count $S + 500$; attempt close.

Expected: Behaviour matches FR text (flag for investigation and/or allowed close with reason); always audited.

Refs: FR-16, FR-17.

TC-OFJ-04 P0

CBS export stub after successful OFJ

Objective: Closed day triggers export toward middleware/mock.

Preconditions: TC-OFJ-01 success path available.

Steps: 1) Complete balanced OFJ. 2) Observe export job / mock endpoint / file. 3) Re-trigger export.

Expected: Payload received once per logical day export; second call idempotent (no duplicate ledger effect).

Refs: FR-18.

TC-CLI

TC-CLI-01 P0

Client views own profile only

Objective: Authenticated member sees only their profile.

Preconditions: Client user for C01; another member C02 exists.

Steps: 1) Login as C01. 2) Open profile. 3) Attempt access to C02 profile URL/id if exposed.

Expected: C01 data shown; C02 access denied or not reachable.

Refs: FR-20.

TC-CLI-02 P0

Client views balance after sync

Objective: Balance reflects posted collections.

Preconditions: Known collection for C01 synced to server.

Steps: 1) Login as C01. 2) Open balance view. 3) Compare with back-office/server total for C01.

Expected: Amounts match within defined rounding (integer XAF).

Refs: FR-21.

TC-CLI-03 P1

Contribution history list

Objective: Client can list past contributions.

Preconditions: C01 has ≥ 2 synced collections; C03 has zero.

Steps: 1) Login C01 — open history. 2) Login C03 — open history.

Expected: C01 sees ordered list (newest first); C03 sees empty state message.

Refs: FR-22.

TC-SEC

TC-SEC-01 P0

Protected API rejects missing JWT

Objective: Unauthenticated callers cannot collect or read private data.

Preconditions: Gateway/Core running; collection endpoint known from Swagger.

Steps: 1) `POST /api/v1/collections` without Authorization header. 2) Repeat with garbage token.

Expected: HTTP 401; no row created.

Refs: FR-01 (session), System Design security.

TC-SEC-02 P0

Broken object-level authorization — cross-branch deny

Objective: Agent of branch A cannot read or modify branch B resources.

Preconditions: Agent A1 bound to BR-YDE-01; resource IDs exist for branch BR-DLA-02 (seed or fixture).

Steps: 1) Login as A1; capture JWT. 2) GET escrow or collections for a BR-DLA-02 agent/client id. 3) Attempt POST collection attributed to the other branch.

Expected: 403 Forbidden or empty/not-found without data leak; no write side-effect.

Refs: STRIDE Elevation / BOLA; System Design.

TC-SEC-03 P1**Gateway rate limiting**

Objective: Burst traffic receives 429 before Core is overwhelmed.

Preconditions: Rate limit configured (e.g. N req/min on login or public route).

Steps: 1) Script or repeat requests above threshold from one IP. 2) Observe status codes.

Expected: Excess calls return 429; legitimate spaced calls still succeed after window.

Refs: System Design — DoS control.

TC-SEC-04 P0**Offline database not readable as cleartext**

Objective: Local collection/tracking store is encrypted at rest.

Preconditions: At least one offline collection stored; device file access (debug build).

Steps: 1) Locate app DB file. 2) Open without key as raw bytes/text. 3) Search for known client name or amount string.

Expected: No plaintext match; content appears ciphertext/unreadable.

Refs: NFR-01, STRIDE Disclosure/Tampering.

TC-UX — Usability smoke**TC-UX-01 P0****Agent completes one deposit under five minutes**

Objective: Primary field path is efficient for non-expert agents.

Preconditions: A1 logged in; GPS on; client C01 known; escrow sufficient.

Steps: 1) Start timer. 2) Identify C01. 3) Enter amount 2,000 XAF and matching denomination. 4) Confirm. 5) Stop timer on success feedback.

Expected: End-to-end ≤ 5 minutes on first assisted attempt; no dead-end screens; success feedback visible.

Refs: NFR-06 (inclusive ergonomics); Test Plan usability smoke type.

Traceability (MVP and referenced FRs)

FR/BR	Topic	Cases	P
FR-01	SSO / IMEI	TC-AUTH-01..03, TC-SEC-01	P0
FR-02	Enrollment	TC-AUTH-04	P0
FR-03	Escrow top-up (MVP path)	TC-ESC-04	P1
FR-04 / BR-03	Escrow lockout + boundaries	TC-ESC-01..03, TC-ESC-06..08	P0
FR-05	Ceiling override	TC-ESC-05	P1
FR-06	Client identification	TC-COL-05	P0
FR-07	Offline mode	TC-OFF-01..04, TC-ESC-03	P0
FR-08 / BR-02	Denomination	TC-COL-01, TC-COL-04	P0
FR-10	Trail geolocation	TC-GEO-01..02	P0
FR-11	Route / map	TC-GEO-03	P1
FR-12 / BR-05	GPS gate / geotag	TC-COL-02,03,06	P0
FR-16 / BR-01	OFJ reconcile	TC-OFF-01..03	P0
FR-17	Variance debt	TC-OFF-02	P0
FR-18	CBS export stub	TC-OFF-04	P0
FR-20	Client profile	TC-CLI-01	P0
FR-21	Client balance	TC-CLI-02	P0
FR-22	History	TC-CLI-03	P1
NFR-01	AES-256 offline	TC-OFF-01, TC-SEC-04	P0
NFR-06	Ergonomics	TC-UX-01	P0
NFR-08	Sync behaviour	TC-OFF-02..04	P0
NFR-09	Battery pulse	TC-GEO-01	P0
NFR-10	Off-duty privacy	TC-GEO-04	P1
STRIDE	BOLA / DoS	TC-SEC-02, TC-SEC-03	P0/P1

Table 1: Requirement-to-test traceability. MVP P0 rows support the exit criterion “every MVP FR mapped”. FR-09, FR-13–15, FR-19 remain V1.1/V2 (no P0 case required for freeze).

5. Test Procedure Specification

Identifier

TPS-MICROFI-V1.0

Purpose

Define the **sequence** in which related test cases are executed so setup is shared and dependencies are respected.

Procedure PROC-SMOKE — Build smoke (30 min)

1. Transmit build (Section 6).
2. Start Docker stack; open Swagger; GET /health.
3. TC-AUTH-01 on seed agent A1.

4. TC-COL-01 one online collection.
5. Log results in Test Log.

Procedure PROC-MONEY — Money path (P0)

Order: TC-AUTH-01 → TC-ESC-01 → TC-ESC-08 → TC-ESC-06 → TC-ESC-07 → TC-COL-01 → TC-COL-04 → TC-ESC-02 → TC-COL-02.

Special: Reset escrow between boundary ESC cases; use remaining = 10,000 XAF fixture for TC-ESC-06..08.

Procedure PROC-OFFLINE — Offline & sync (P0)

Order: TC-AUTH-01 → TC-OFF-01 → TC-OFF-03 → TC-OFF-02 → TC-OFF-04 → TC-ESC-03.

Special: Airplane mode; do not clear app data mid-procedure.

Procedure PROC-OFJ — End of day (P0)

Order: Ensure day's collections synced → TC-OFJ-01 → TC-OFJ-02 → TC-OFJ-04.

Procedure PROC-GEO — Trail (P0/P1)

Order: Start field session → TC-GEO-01 (wait ≥ 15 min offline) → TC-GEO-02.

Procedure PROC-ACCEPT — Acceptance demo

Duration ≈ 15 –20 min on physical Android.

1. Login A1; reject wrong PIN once (AUTH-01/02).
2. Online collection with denomination + GPS (COL-01).
3. GPS off → blocked (COL-02).
4. Escrow lockout (ESC-02).
5. Offline two collections; sync once; no duplicates (OFF).
6. OFJ balanced close; optional negative variance (OFJ).
7. Client profile + balance (CLI).

Pass: All steps without Critical defect; GPS gate and lockout not bypassable.

6. Test Item Transmittal Report

Use this form whenever development hands a build to testing (or self-test for academic solo work).

Field	Value
Report ID	TITR-MICROFI-____
Date / time	
Transmitter (dev)	
Receiver (test)	
Build / commit / tag	
Artifacts	APK / Docker images / OpenAPI revision
Environment	Local / Staging
Features included	(e.g. FR-07 offline sync)
Known limitations	
Smoke status before handoff	Pass / Fail
Location of binaries	
Signature transmitter	
Signature receiver	

Rule: No acceptance procedure starts without a completed transmittal for the build under test.

7. Test Log

Identifier

TL-MICROFI-[date]

Purpose

Chronological record of who ran which procedure/case, when, on which build, and the result.

Date/Time	Tester	Build	Proc/TC	Result	Notes
				Pass / Fail	
				Pass / Fail	
				Pass / Fail	
				Pass / Fail	

Table 2: Test Log template — duplicate rows as needed.

Status values: Pass, Fail, Blocked, Not Run, Skipped.

8. Test Incident Report

File one report per deviation from expected results. Link to Test Log entry and affected TC/FR.

Field	Value
Incident ID	INC-MICROFI-____
Date detected	
Tester	
Build / commit	
Procedure / TC	
Related FR/BR	
Severity	Critical / Major / Minor / Trivial
Summary	
Steps to reproduce	
Expected result	
Actual result	
Environment	Device, OS, network mode
Attachments	Logs, screenshots
Status	New / Confirmed / Fixed / Verified / Closed
Assignee	

Severity guide

Critical: wrong amount, double post, lockout bypass, collection without GPS.
Major: sync loss, wrong OFJ delta, crash on main path.
Minor / Trivial: UI wording, cosmetic.

9. Test Summary Report

Identifier

TSR-MICROFI-V1.0 — completed at MVP freeze / acceptance.

Summary

Field	Entry
Project / version	MICROFI Collect V1.0 MVP
Test period	From ____ to ____
Build(s) tested	
Procedures executed	PROC-SMOKE / MONEY / OFFLINE / OFJ / GEO / ACCEPT
P0 cases planned	
P0 passed / failed / blocked	
Incidents opened / closed	
Critical open at freeze	0 required for acceptance
Environment	

Evaluation

- Features that met pass criteria: _____
- Features deferred: _____
- Residual risks: GPS device dependency; CBS mock only; battery NFR qualitative.

Summary of results

(Narrative: overall quality judgment, readiness for academic demo / pilot.)

Recommendations

1. Accept / conditional accept / reject MVP freeze.
2. Priority fixes before pilot: _____
3. Next test cycle focus (V1.1): FR-05, FR-09, FR-13, FR-19, FR-22.

Approvals

Author: _____ Supervisor: _____ Date: _____

10. Automation Notes

- **Backend:** JUnit / WebTestClient for escrow, denomination, GPS filter, idempotent sync.
- **API:** Swagger UI / OpenAPI (springdoc) for interactive contract testing; include negative cases.
- **Mobile:** Queue persistence tests; GPS on physical device.
- **CI:** Unit + API smoke on merge; full device suite before acceptance procedure.

MICROFI Collect — PK Fokam Research Center / ENSPY 4GI

Test Documentation v1.2 — Full IEEE 829-style pack

Design system aligned with Business Analysis v3.1